FIITJEE

JEE(Main)-2018

ALL INDIA TEST SERIES

ANSWERS, HINTS & SOLUTIONS

FULL TEST – VII
(Main)

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Physics

PART – I

SECTION-A

1. Given $v = a_0t$

$a_t = \frac{dv}{dt} = a_0$

Now, $v^2 = u^2 + 2a_tS$

$= 0 + 2a_0\frac{\pi R}{2}$

$v^2 = \pi a_0 R$

So, $a_n = \frac{v^2}{R} = \pi a_0$

The angle between the velocity vector and the acceleration vector is

$\phi = \tan^{-1}\left(\frac{a_n}{a_t}\right) = \tan^{-1}\left(\frac{\pi a_0}{a_0}\right) = \tan^{-1}(\pi)$

2. The two blocks will move with a common acceleration

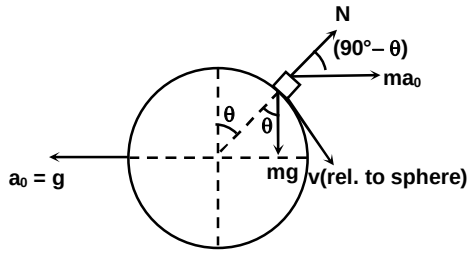
$a = \frac{100-30}{10} = 7\ m/s^2$

Now, for the block 'A'

$20 - f_S = 2 \times 7$

$\therefore f_S = 6N$

3. Let v is the velocity of the body relative to the sphere when it loses contact with the sphere.



$mg\cos\theta - N - ma_0\sin\theta = \frac{mv^2}{R}$

To lose contact with the sphere, N = 0

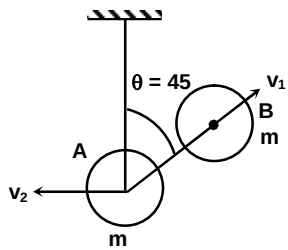
$mg\cos\theta - ma_0\sin\theta = \frac{mv^2}{R}$

Also, $mgR(1-\cos\theta) + ma_0R\sin\theta = \frac{1}{2}mv^2$

$mg = \frac{3mv^2}{2R}$

$\therefore\ v = \sqrt{\frac{2gR}{3}}$

4. Using conservation of momentum of the system along horizontal direction



$\frac{mv_0}{\sqrt{2}} = mv_2 - \frac{mv_1}{\sqrt{2}}$

$\sqrt{2}v_2 - v_1 = v_0$ …(i)

$v_2\cos 45° + v_1 = v_0$

$\frac{v_2}{\sqrt{2}} + v_1 = v_0$

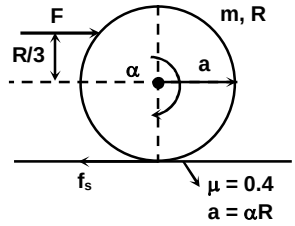
$\sqrt{2}v_2 + 2v_1 = 2v_0$ ..(ii)

Solving (i) and (ii), we get

$3v_1 = v_0$

$v_1 = \frac{v_0}{3} = \frac{6}{3} = 2$ m/s

5. $F - f_s = ma$ …(i)



$\frac{FR}{3} + f_sR = \frac{2}{3}mR^2\alpha$

$\frac{F}{3} + f_s = \frac{2}{3}ma$ [since, $a = R\alpha$] …(ii)

Solving (i) and (ii), we get

$\frac{4F}{3} = \frac{5}{3}ma$

$a = \frac{4F}{5m}$

Now, from equation (i)

$f_S = F - ma = F - \frac{4F}{5} = \frac{F}{5}$

Hence, $f_S \le \mu N$

$F/5 \le \mu mg$

$F \le 5\ \mu mg$

$\therefore F_{max} = 5 \times 0.4 \times 20 = 40$ N

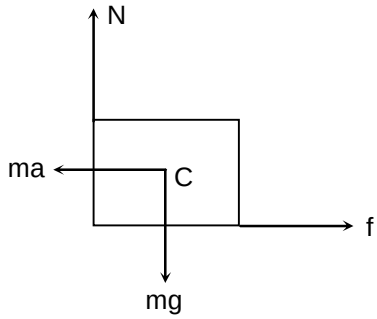
6. For toppling of the block

$ma\frac{h}{2} > mg\frac{w}{2}$

$a > \frac{w}{h}g$

$a > \frac{15}{60} \times 10$

$a > 2.5\ m/s^2$



7. $N = mg\cos\theta$

$f_k = \mu N = 2\tan\theta mg\cos\theta = 2mg\sin\theta$

$f_k - mg\sin\theta = ma$

$2mg\sin\theta - mg\sin\theta = ma$

$\therefore \quad a = g\sin\theta$

Now, $\tau_{cm} = I_{cm}\alpha$

$f_kR = \frac{2}{5}mR^2\alpha$

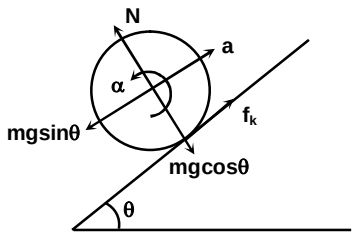
$2mg\sin\theta = \frac{2}{5}mR\alpha$

$\alpha = \frac{5g\sin\theta}{R}$

When the rolling without slipping starts

$v = \omega R$

$v_0 + at = (\omega_0 - \alpha t)R$

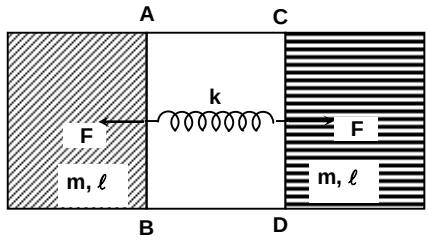


$v_0 + g\sin\theta t = \omega_0 R - 5g\sin\theta t$

$v_0 + 6g\sin\theta t = 2v_0$

$\therefore \quad t = \dfrac{v_0}{6g\sin\theta}$

8. Force of surface tension on each wire is $F = 2T\ell$



$\dfrac{md^2x}{dt^2} = F - 2Kx$

$= -2K\left(x - \dfrac{F}{2k}\right)$

$\dfrac{d^2x}{dt^2} = -\dfrac{2k}{m}\left(x - \dfrac{F}{2k}\right)$

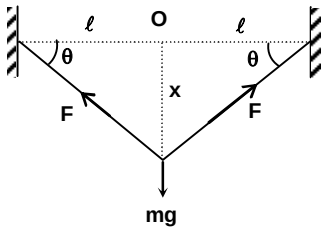
$\therefore$ Amplitude of SHM,

$A = \dfrac{F}{2k} = \dfrac{2T\ell}{2k} = \dfrac{T\ell}{k}$

$\therefore$ Maximum elongation produced in the spring is

$\Delta\ell_{max} = 4A = \dfrac{4T\ell}{k}$

9. $2F\sin\theta = mg$



$F = \dfrac{mg}{2\sin\theta} = \dfrac{mg\ell}{2x}$ [$\because$ θ is small]

$\Delta\ell = \left(\sqrt{\ell^2 + x^2} - \ell\right)$

$= \ell\left[\left(1 + \dfrac{x^2}{\ell^2}\right)^{1/2} - 1\right]$

$\Delta\ell = \ell\left[1 + \dfrac{1}{2}\dfrac{x^2}{\ell^2} - 1\right]$ [$\because$ $x << \ell$]

$\therefore \quad \dfrac{\Delta\ell}{\ell} = \dfrac{x^2}{2\ell^2}$

Now $\dfrac{F}{A} = Y\dfrac{\Delta\ell}{\ell}$

$F = AY\dfrac{\Delta\ell}{\ell}$

$\dfrac{mg\ell}{2x} = AY\left(\dfrac{x^2}{2\ell^2}\right)$

$x^3 = \dfrac{mg\ell^3}{AY}$

$\therefore \quad x = \ell\left(\dfrac{mg}{AY}\right)^{1/3}$

10. Using conservation of energy

$\dfrac{1}{2}mv^2 = -\dfrac{GMm}{R} - \left(\dfrac{-11GMm}{8R}\right)$

$$\frac{1}{2}mv^2 = \frac{3GMm}{8R}$$

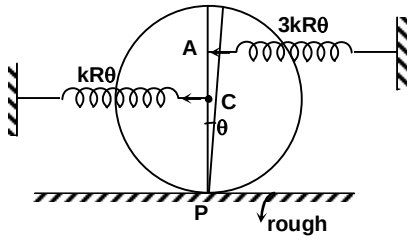
$$\therefore\ v = \sqrt{\frac{3GM}{4R}}$$

11. $$\frac{3mR^2}{2}\frac{d^2\theta}{dt^2} = -kR^2\theta - 3kR\theta\left(\frac{3R}{2}\right)$$

$$\frac{3mR^2}{2}\frac{d^2\theta}{dt^2} = -\frac{11kR^2\theta}{2}$$

$$\frac{d^2\theta}{dt^2} = -\left(\frac{11K}{3m}\right)\theta$$

$\therefore$ Time Period, $T = 2\pi\sqrt{\frac{3m}{11k}}$



12. Given, VT = constant

PV^2 = constant

Polytropic constant, x = 2

$\therefore$ Molar heat capacity,

$$C = C_V + \frac{R}{1-x}$$

$$= \frac{3R}{2} + \frac{R}{1-2}$$

$$= \frac{R}{2}$$

$$\therefore\ \Delta Q = nC\Delta T = 2\times\frac{R}{2}\times 300 = 300R$$

$$\Delta U = nC_v\Delta T = 2\times\frac{3R}{2}\times 300 = 900R$$

$$\therefore\ \frac{\Delta U}{\Delta Q} = \frac{900R}{300R} = 3$$

13. Efficiency of carnot engine is, $\eta = 1 - \frac{T_2}{T_1}$

$$0.1 = 1 - \frac{T_2}{T_1}$$

$$\therefore\ \frac{T_2}{T_1} = 0.9$$

Coefficient of performance (COP) of a refrigerator = $\frac{T_2}{T_1 - T_2} = \frac{0.9T_1}{T_1 - 0.9T_1} = 9$

$\therefore$ COP = 9

14. The frequency of vibration of air column in the pipe is $f_1 = \frac{3v}{4\ell} = \frac{3\times 340}{4\times 0.85} = 300$ Hz.

The frequency of vibration of string is $f_2 = f_1 + 6 = 300 + 6 = 306$ Hz

Now, $f_2 = \frac{3}{2\ell}\sqrt{\frac{F}{\mu}}$

$$306 = \frac{3}{2\times 0.5}\sqrt{\frac{F\times 0.5}{2.5\times 10^{-3}}}$$

$$306 = 30\sqrt{2F}$$

$$\sqrt{2F} = 10.2 \quad \therefore \quad F = 52\ N$$

15. $f_1 = \left(\frac{v - v_D}{v - v_s}\right) f$ and the frequency of sound received by the observer after reflection from the stationary wall is $f_2 = \left(\frac{v + v_D}{v - v_s}\right) f$

$\therefore$ Beats frequency received by the observer, $f_b = f_2 - f_1 = \left(\frac{v + v_D}{v - v_s}\right) f - \left(\frac{v - v_D}{v - v_s}\right) f$

$$\therefore \quad f_b = \frac{(2v_D) f}{(v - v_s)} = \frac{2\times 2}{(330 - 10)} \times 320 = 4\,Hz.$$

16. The electric field strength at point 'P' is

$$E = E_1 - E_2$$

$$= \frac{\rho R}{4\varepsilon_0} - \frac{\rho R}{54\varepsilon_0}$$

$$E = \frac{25\rho R}{108\varepsilon_0}$$

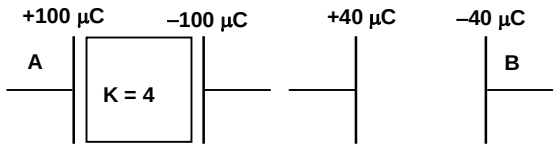
17. $Q_A = 1\times 100 = 100\mu C$

$Q_B = 2\times 20 = 40\mu C$

When the two capacitors are connected with opposite polarities.

$q_A = 40\mu C,\ q_B = 20\mu C$

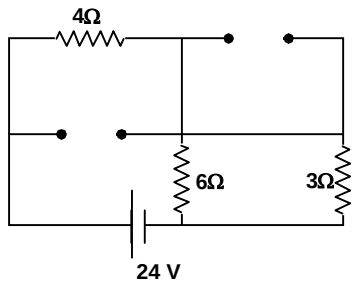
The heat dissipated after the two capacitors connected



$$\Delta H = U_i - U_f = \left[\frac{(100)^2}{2\times 4} + \frac{(40)^2}{2\times 2}\right] - \frac{(60)^2}{2\times 6}$$

$$= (1250 + 400) - 300 = 1350\,\mu J = 1.35\times 10^{-3} J$$

18. In steady state, the equivalent circuit is shown.



$$R_{eq.} = 4 + \left(\frac{6\times 3}{6 + 3}\right)$$

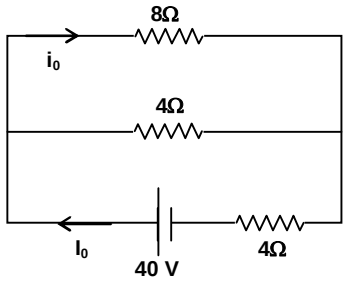
$$R_{eq} = 4 + 2 = 6\Omega$$

$$\therefore \quad I_S = \frac{\varepsilon}{R_{eq}} = \frac{24}{6} = 4A\ .$$

19. Time constant, $\tau = R_{eq}C_{eq}$

$= 10 \times 20 \times 10^{-6}$

$= 0.2 \times 10^{-3}$ sec



The current through the battery just after closing the switch is

$I_0 = \frac{40 \times 3}{20} = 6A$

The current through R = 8 Ω register at t = 0 is

$i_0 = \frac{1}{3} \times I_0 = \frac{1}{3} \times 6 = 2A$

The current through R = 8Ω resistor at time 't'

$i = i_0 e^{-t/\tau}$

$i = 2\, e^{-t/0.2\times10^{-3}}$

$\therefore$ at $t = 0.4 \times 10^{-3}$ sec.

$\ell = 2e^{-2} = 2 \times 0.135 = 0.27\,A$

20. $\tau = \frac{\sqrt{3}}{4} Bi\ell^2$

Now, $\int \tau dt = I\omega$

$\frac{\sqrt{3}}{4} B\ell^2 \int i dt = \frac{5m\ell^2\omega}{4}$

$\frac{\sqrt{3}}{4} B\ell^2 Q = \frac{5m\ell^2}{4}\omega$

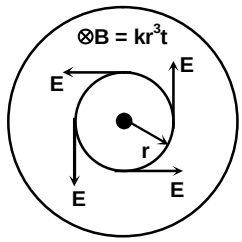
$\therefore \quad \omega = \frac{\sqrt{3}BQ}{5m}$

21. The net magnetic field intensity at the centre 'O' is

$\vec{B}_0 = \frac{\mu_0 I}{4\pi R}\left(-\hat{k}\right) + \frac{\mu_0 I}{4R}\left(-\hat{i}\right)$

$\therefore \quad B_0 = \frac{\mu_0 I}{4\pi R}\sqrt{1+\pi^2}$

22. Using Faraday's law



$E 2\pi r = -\frac{d}{dt}\left[-\int_0^r B 2\pi r dr\right]$

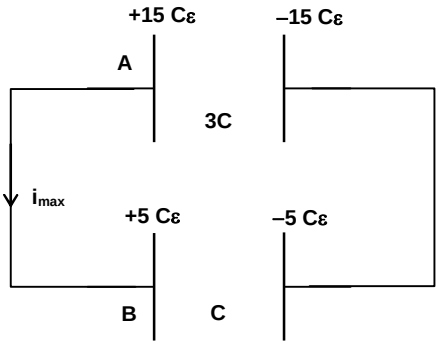
$= \frac{d}{dt}\left[\int_0^r Kr^3 t 2\pi r dr\right]$

$E 2\pi r = \frac{d}{dt}\left(2\pi kt \frac{r^5}{5}\right)$

$E = \frac{Kr^4}{5}$, when $r \le R$

$\therefore$ at $r = \frac{R}{2}$, $\quad E = \frac{KR^4}{80}$

23. When the current through the inductor is maximum,



$V_L = L\dfrac{di}{dt} = 0$

Now, using energy conservation

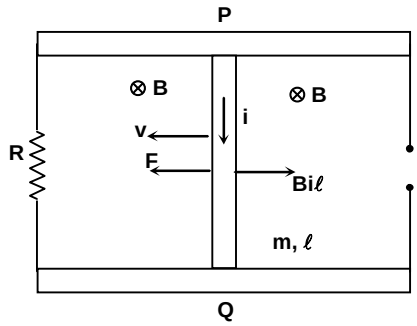
$\dfrac{1}{2}Li_{max}^2 = \left(\dfrac{1}{2}\times 3C\times 16\varepsilon^2 + \dfrac{1}{2}C\times 64\varepsilon^2\right) - \dfrac{1}{2}(3C+C)25\varepsilon^2$

$\dfrac{1}{2}Li_{max}^2 = 56C\varepsilon^2 - 50C\varepsilon^2$

$\dfrac{1}{2}Li_{max}^2 = 6C\varepsilon^2$

$\therefore \quad i_{max} = \varepsilon\sqrt{\dfrac{12C}{L}}$

24. When the terminal velocity is acquired



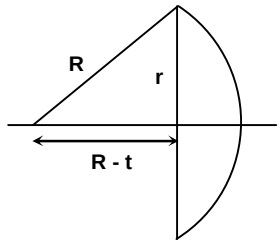
$i = \dfrac{Bv\ell}{R}$

$F = Bi\ell$

$F = \dfrac{B^2\ell^2 v}{R}$

$\therefore \quad v = \dfrac{FR}{B^2\ell^2}$

25. $R^2 = (R-t)^2 + r^2$



$R^2 = R^2 + t^2 - 2Rt + r^2$

$r^2 = 2Rt$ (t^2 is neglected)

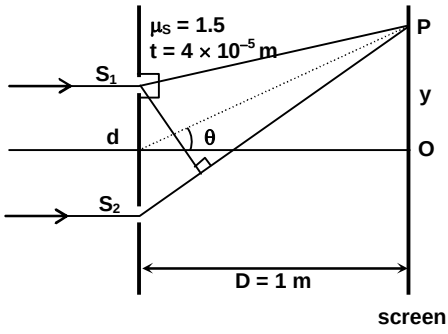
$R = \dfrac{r^2}{2t} = \dfrac{4\times 4}{2\times 0.4} = 20$ cm

R = 20 cm

$\therefore \quad \dfrac{1}{f} = (1.5-1)\left(\dfrac{1}{\infty} - \dfrac{1}{-20}\right)$

$\dfrac{1}{f} = \dfrac{0.5}{20}$ $\therefore$ f = + 40 cm

26. For central maxima $d\sin\theta = (\mu_s - 1)t$



$1\times 10^{-3}\sin\theta = (1.5-1)4\times 10^{-5}$

$1\times 10^{-3}\sin\theta = 2\times 10^{-5}$

$\sin\theta = \dfrac{1}{50}$

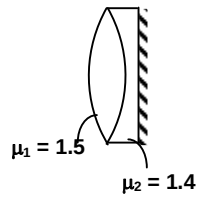
$\tan\theta = \dfrac{1}{50}$ (for small 'θ')

$\dfrac{y}{D} = \dfrac{1}{50}$

$\therefore \quad y = \dfrac{D}{50} = \dfrac{100}{50}\text{cm} = 2\text{ cm}\ (\text{above point 'O'})$

27. $\frac{1}{f_1} = (1.5-1)\left(\frac{1}{R}+\frac{1}{R}\right) = \frac{1}{R}$

$\therefore \quad f_1 = R$



$\frac{1}{f_2} = (1.4-1)\left(\frac{1}{-R}-\frac{1}{\infty}\right)$

$\frac{1}{f_2} = -\frac{0.4}{R}$

$-\frac{1}{F} = \frac{2}{f_1}+\frac{2}{f_2}-\frac{1}{f_m}$

Since image coincides with the object,

F = – 10 cm

$\frac{-1}{-10} = \frac{2}{R}+2\left(\frac{-0.4}{R}\right)-\frac{1}{\infty}$

$\frac{1}{10} = \frac{2}{R}-\frac{0.8}{R}$

$\frac{1}{10} = \frac{1.2}{R}$

$\therefore$ R = 12 cm

28. Energy of each photon

$E = \frac{hC}{\lambda} = \frac{1240}{400} = 3.1\,eV$

Now, $K_{max} = E-\phi$

= 3.1 – 2.3

$\therefore \quad K_{max} = 0.8$ eV

Energy of He^+-ion in third excited state

$E_4 = -13.6\times\frac{4}{(4)^2} = -3.4$ eV

Now, the energy of the photon emitted during combination,

$\Delta E = K_{max} - E_4$

$= 0.8-(-3.4)$

= 4.2 eV

29. The intensity of the light after passing through the first Polarised = $\frac{I_0}{2}$. Now, the intensity of the light after getting passed through the second Polaroid will be

$I = \frac{I_0}{2}\cos^2\theta$

$= \frac{I_0}{2}\cos^2 45^\circ = \frac{I_0}{4}$

30. The pitch of the screw gauge, P = 1 mm

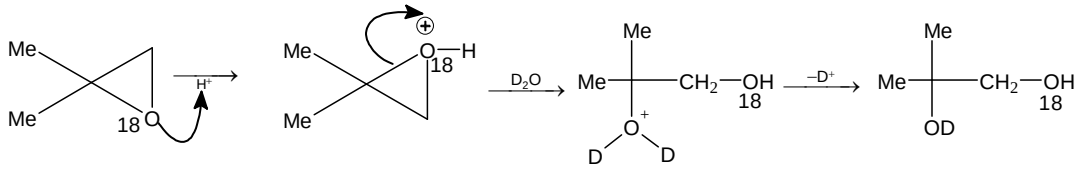
Least count, L.C. = $\frac{P}{N} = \frac{1\,mm}{100} = 0.01\,mm$

$\therefore$ Diameter of the wire, d = 2mm + 58 × L.C.
= 2mm + 58 × 0.01 mm
= 2.58 mm

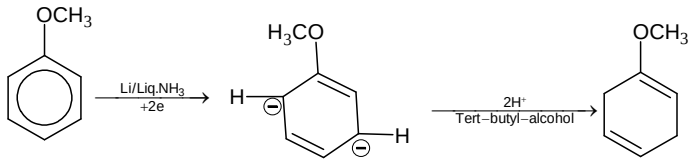
Chemistry PART – II

SECTION-A

32.



33.



1-methoxy cyclohexa-1,4-diene

36. $10^{13} e^{-\frac{Ea_1}{RT}} = 10^{14} e^{-\frac{Ea_2}{RT}}$

T = 230 K

37. Orbital angular momentum per electron $= \sqrt{\ell(\ell+1)}\,\frac{h}{2\pi}$

$\sqrt{2(2+1)}\,\frac{h}{2\pi}$

$\sqrt{6}\,\hbar$



For Fe^{3+} :

$\therefore$ Total orbital angular momentum in Fe^{3+} salt (per molecule) $= 5\sqrt{6}\,\hbar$

40. $\frac{r_+}{r_-} = 0.414$ to 0.732 (C. N. = 6)

$\frac{r_+}{r_-} = 0.225$ to 0.414 (C. N. = 4)

Hence option (A) correct.

41. $4Fe^{3+} + K_4\left[Fe(CN)_6\right] \longrightarrow Fe_4\left[Fe(CN)_6\right]_3 + 4K^+$

42. According to Hardy-Schulze law coagulation power $\propto$ valency of the counter ion.

43. For triatomic gas

$\gamma = 1.33$, $C_P = 4R$

$T_2 = T_1\left(\frac{V_1}{V_2}\right)^{\gamma-1}$

$= 300\left(\frac{5}{40}\right)^{\frac{1}{3}}$

$= 300\left(\frac{1}{8}\right)^{\frac{1}{3}} = 300 \times \frac{1}{2} = 150$

$\Delta H = n \times C_P (T_2 - T_1)$

$= 3 \times 4 \times 8.314 (150 - 300)$

$= -14.965$ kJ

44. x is Borax

$Na_2B_4O_7.10H_2O$

$Na_2B_4O_7 + 7H_2O \longrightarrow 2NaOH + 4H_3BO_3$

$Na_2B_4O_4 \longrightarrow \underbrace{2NaBO_2 + B_2O_3}_{\text{Glassy bead}}$

$Na_2B_4O_7 + H_2SO_4 + 5H_2O \longrightarrow Na_2SO_4 + 4H_3BO_3$

45. $\frac{P_1V_1}{T_1} = \frac{2P_1 \times 2V_1}{T_2}$

$T_2 = 4T_1$

$4T_1 \times n_1 = T_2'n_2'$

$4T_1n_1 = T_2' \times \frac{5}{4}n_1 \quad \left(n_2 = n_1 + \frac{n_1}{4}\right)$

Therefore $T_2' = \frac{16}{5}T_1$

46. Gold in $AuCl_4^-$ has to be reduced at an electrode with excess electron. Hence, stirring is essential.

47. Hoffman Bromide reaction.

48. $CaF_2 \rightleftharpoons Ca^{2+} + 2F^-$

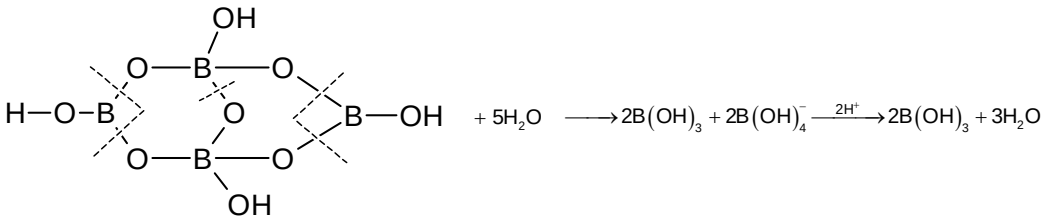
$K_{SP} = 4S^3 = 4 \times 10^{-11}$

$\Rightarrow S = 2.15 \times 10^{-4}$ M

Amount of F^- in 20,000 L Sol. = $2 \times 20{,}000 \times 2.15 \times 10^{-4} = 8.6$ mole

49. $\overset{\delta+}{Cl}\overset{\delta-}{F}$

54.



57. Moles of oxalic acid $= \frac{50 \times 1}{1000} = \frac{1}{20}$

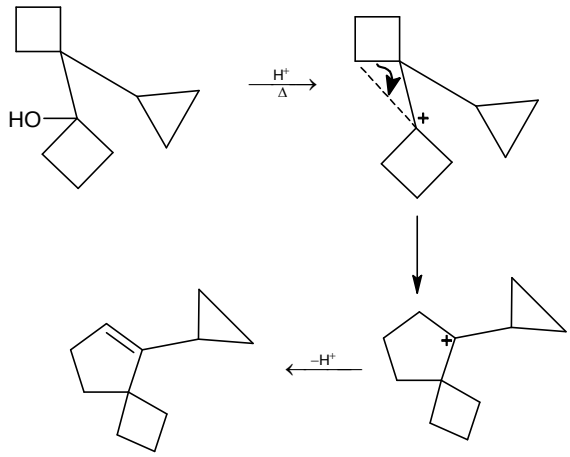
Weight of oxalic acid $= \frac{1}{20} \times 126 = 6.3$ gm

Conc. of solution after adsorption = 0.4 M

Moles of oxalic acid after adsorption $= \frac{50 \times 0.4}{1000} = \frac{1}{50}$

Wt. of oxalic acid left after adsorption $= \frac{1}{50} \times 126 = 2.52$ gm

60.



Mathematics PART – III

SECTION-A

61. Let $\log x = t$

$\frac{dt}{dx} = \frac{1}{x} \Rightarrow dx = x\,dt;\ dx = e^t\,dt$

$\int \frac{\left(t^2 \tan^{-1} t + 1 + \tan^{-1} t\right)}{1+t^2} e^t dt$

$\int \frac{\tan^{-1} t\left(1+t^2\right)+1}{\left(1+t^2\right)} e^t dt = \int e^t \left(\tan^{-1} t + \frac{1}{1+t^2}\right) dt = e^t \tan^{-1} t + c = x \tan^{-1}(\log x) + c$

62. $\int \frac{11x - 18}{x^{13}\sqrt[6]{1 - \frac{2}{x^{11}} + \frac{3}{x^{12}}}} dx = \int \frac{\frac{11}{x^{12}} - \frac{18}{x^{13}}}{\sqrt[6]{1 - \frac{2}{x^{11}} + \frac{3}{x^{12}}}} dx$

Let $1 - \frac{2}{x^{11}} + \frac{3}{x^{12}} = t$

$\Rightarrow \frac{dt}{dx} = \frac{22}{x^{12}} - \frac{36}{x^{13}}$

$\Rightarrow \frac{1}{2}\int \frac{1}{t^{\frac{1}{6}}} dt$

$\Rightarrow \frac{1}{2} \times 6 \frac{t^{\frac{5}{6}}}{5} + c$

$\Rightarrow \frac{3}{5}\left(1 - \frac{2}{x^{11}} + \frac{3}{x^{12}}\right)^{\frac{5}{6}} + c = \frac{3}{5}\frac{\left(x^{12} - 2x + 3\right)^{\frac{5}{6}}}{x^{10}} + c$

63. $g(x) = \int \frac{e^{\cos x}}{x}$

$I = \int_1^2 \frac{3e^{\cos x^3}}{x} dx$

Let $x^3 = t \Rightarrow \frac{dt}{dx} = 3x^2$

$\Rightarrow \int \frac{3e^{\cos t}}{3x^3} dt \Rightarrow \int_1^8 \frac{e^{\cos t}}{t} dt = g(8) - g(1) = 10a_3 - 3a_1$

64. $PQ = \text{diameter} = \frac{|10-(-6)|}{\sqrt{2}} = 8\sqrt{2}$

$\Rightarrow$ Radius of circle is $4\sqrt{2}$

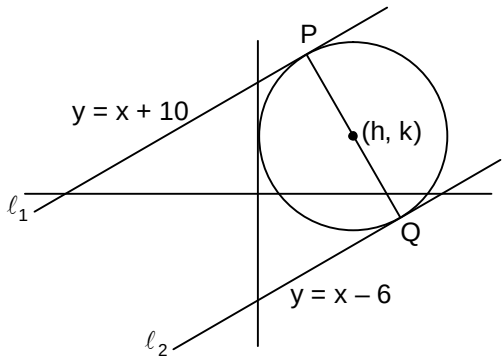
Centre lies on the line parallel to ℓ_1 and ℓ_2, i.e. it lies on the line y = x + b and at a distance of $4\sqrt{2}$ from ℓ_1 and ℓ_2 $\therefore$ b = 2

i.e. centre lies on y + x = 2

$\therefore$ Let centre be $(\alpha, \alpha + 2)$ as it touches the y-axis, α = radius $\therefore$ $\alpha = 4\sqrt{2}$

$\therefore$ Centre is $\left(4\sqrt{2}, 4\sqrt{2}+2\right)$

h + k = $2+8\sqrt{2}$ $\therefore$ a + b = 10



65. Let the first three terms of an A.P. be a, a + d, a + 2d then $(a + d + 1)^2 = (a + 1)(a + 2d + 7)$

$\Rightarrow d^2 = 6a + 6 \Rightarrow \frac{d^2}{6} = a + 1$

Common ratio is $\frac{a+d+1}{a+1} = \frac{(a+1)+d}{a+1} = \frac{\frac{d^2}{6}+d}{\frac{d^2}{6}} = \frac{d+6}{d}$ for common ratio to be an integer d is equal to 1, 2, 3, 6

66. As line is perpendicular to normal of the plane

$\Rightarrow$ 3a + 2b + 12 = 0 and as (2, –1, 3) lies in the plane

$\Rightarrow$ 2a – b + 9 = 8

$\Rightarrow$ 2a – b + 1 = 0

On solving we get a = –2, b = –3

67. $2x^2 y\,dx - x^3\,dy = y^2\,dx$

$\Rightarrow \frac{2xy\,dx - x^2 dy}{y^2} = \frac{dx}{x}$

$\Rightarrow \int d\left(\frac{x^2}{y}\right) = \int \frac{dx}{x}$

$\Rightarrow y = \frac{x^2}{\log x + c}$

It passes through (1, 1)

$\Rightarrow$ c = 1

$\therefore y = \frac{x^2}{\log x + 1}$

$\therefore$ at x = 2; $y = \frac{4}{\log 2 + 1} = \frac{4}{\log(2e)}$

68. $\frac{1}{\sqrt{\alpha}} + \frac{1}{\sqrt{\beta}} = -\frac{q}{p}$

$\frac{1}{\sqrt{\alpha\beta}} = \frac{1}{p} \Rightarrow \sqrt{\alpha\beta} = p$

$\sqrt{\alpha} + \sqrt{\beta} = -q$

$x^2 - x(3pq - q^3) + p^3 = 0$

Let the roots be α_1, β_1

$\alpha_1\beta_1 = p^3 = (\alpha\beta)^{3/2}$

$\alpha_1 + \beta_1 = 3pq - q^3 = -3\sqrt{\alpha\beta}\left(\sqrt{\alpha} + \sqrt{\beta}\right) + \left(\sqrt{\alpha} + \sqrt{\beta}\right)^3 = \alpha^{\frac{3}{2}} + \beta^{\frac{3}{2}}$

$\Rightarrow \alpha_1 = \alpha^{\frac{3}{2}}$ and $\beta_1 = \beta^{\frac{3}{2}}$

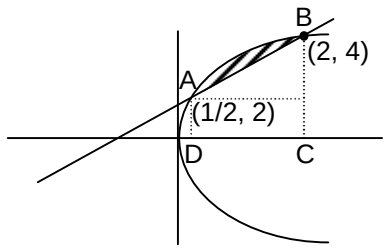
69. $\lim\limits_{x\to 0} f\left(\dfrac{1-\cos 5x}{x^2}\right) = f\left(\lim\limits_{x\to 0}\dfrac{1-\cos 5x}{x^2}\right) = f\left(\lim\limits_{x\to 0}\dfrac{5\sin 5x}{2x}\right) = f\left(\dfrac{25}{2}\right) = 4$

70. $y^2 \le 8x$

$4x - 3y + 4 \le 0$

Area of shaded portion is equal to

$\int\limits_{\frac{1}{2}}^{2}\sqrt{8x}$ – Area of trapezium ABCD



$$2\sqrt{2}\int_{\frac{1}{2}}^{2}\sqrt{x} - \frac{9}{2} = 2\sqrt{2}\left.\frac{x^{\frac{3}{2}}}{\frac{3}{2}}\right|_{\frac{1}{2}}^{2} - \frac{9}{2}$$

$$= \frac{4\sqrt{2}}{3}\left(2\sqrt{2} - \frac{1}{2\sqrt{2}}\right) - \frac{9}{2}$$

$$= \frac{4\sqrt{2}}{3}\left(\frac{8-1}{2\sqrt{2}}\right) - \frac{9}{2}$$

$$= \frac{14}{3} - \frac{9}{2} = \frac{28-27}{6} = \frac{1}{6}$$

71. $4 + 9 - 12 - 30 + k < 0$

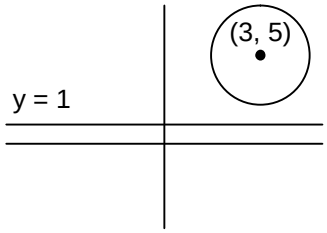
$k < 29$

For the circle to not touch the line y = 1 the radius must be less than 4

$\therefore \sqrt{9 + 25 - k} < 4$

$34 - k < 16$

$k > 18$



72. $2\tan x = \alpha(1 + \tan^2 x)$

$\Rightarrow \dfrac{2\tan x}{1+\tan^2 x} = \alpha \Rightarrow \sin 2x = \alpha$

$\because \angle B$ and $\angle C$ satisfy this equation

$\Rightarrow \sin 2B = \sin 2C$

But angle B > angle C

$\Rightarrow 2C = \pi - 2B$

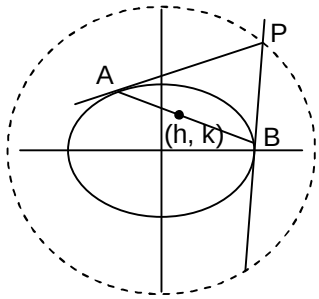
$\Rightarrow B + C = \dfrac{\pi}{2}$

$\therefore \angle A = \dfrac{\pi}{2}$

73. $A^T = 5A + 2I$
Taking transpose both sides $A = 5A^T + 2I$
$\Rightarrow A = 5(5A + 2I) + 2I$
$\Rightarrow 24A + 12I = 0$
$\Rightarrow 2A + I = 0$
$\Rightarrow 2AX + X = 0$
$\Rightarrow 2AX = -X$
$\Rightarrow AX = -\frac{1}{2}X$

74. Converse of $\sim q \rightarrow p$ is $p \rightarrow \sim q$
Its contrapositive statement is $q \rightarrow \sim p$

75. Let the mid-point of chord AB is (h, k)
Then equation of AB is T = S,
$\Rightarrow 49hx + 16ky - 784 = 49h^2 + 16k^2 - 784$
$\Rightarrow 49hx + 16ky = 49h^2 + 16k^2$
Any point on the circle $x^2 + y^2 = 100$ is
P(10 cos θ, 10 sin θ)
Equation of chord of contact AB is
$49x \cdot 10\cos\theta + 16y \cdot 10\sin\theta = 784$
$\because$ Both equation represent equation of same straight line AB



$$\therefore \frac{49h}{490\cos\theta} = \frac{16k}{160\sin\theta} = \frac{49h^2 + 16k^2}{784}$$

$$\cos\theta = \frac{784h}{10\left(49h^2 + 16k^2\right)},\ \sin\theta = \frac{784k}{10\left(49h^2 + 16k^2\right)}$$

$$\therefore \text{Locus } (49h^2 + 16k^2)^2 = \frac{784^2}{100}\left(h^2 + k^2\right)$$

76. $a_i \times b_i$ is a multiple of 3 if atleast one of a_i or b_i is a multiple of 3
If $a_i = 3$ then $b_i = 1, 2, 3, 4, 5, 6, 7$
If $b_i = 3$ or 6 then $a_i = 1, 2, 3, 4$
$\therefore$ Total possibilities = 7 + 4 + 4 – 2 = 13

$$\therefore \text{Probability} = \frac{13}{4C_1 \times 7C_1} = \frac{13}{28}$$

77. $S = {}^{30}C_0 \sin 0x \cos 30x + {}^{30}C_1 \sin x \cos 29x + {}^{30}C_2 \sin 2x \cos 28x + ...$
$+ 30C_{28} \sin 28x \cos 2x + {}^{30}C_{29} \sin 29x \cos x + {}^{30}C_{30} \sin 30x$
$S = {}^{30}C_{30} \sin 30x + {}^{30}C_{29} \sin 29x \cos x + {}^{30}C_{28} \sin 28x \cos 2x + ...$
$+ {}^{30}C_1 \sin x \cos 29x + {}^{30}C_0 \sin 0x \cos 30x$

$$2S = \sum_{r=0}^{30} {}^{30}C_r \left(\sin rx \cos(30 - r)x + \sin(30 - r)x \cos x\right)$$

$$2S = \sum_{r=0}^{30} {}^{30}C_r \sin\left(rx + (30 - r)x\right) = \sum_{r=0}^{30} {}^{30}C_r \sin 30x$$

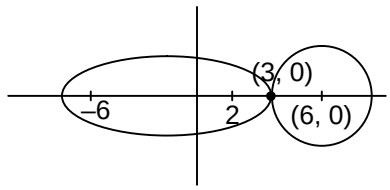
$2S = 2^{30} \sin 30x$
$S = 2^{29} \sin 30x$

78. So just one point is common

$\Rightarrow |z + 6| + |z - 2| = 10$

$\Rightarrow \frac{(x+2)^2}{25} + \frac{y^2}{9} = 1$ ..... (1)

$|z - 6| = 3, \Rightarrow (x - 6)^2 + y^2 = 9$ ..... (2)



79. Slope of tangent to $\frac{dy}{dx} = 2x$

$\therefore$ Slope = 6

Equation of tangent $y - 13 = 6(x - 3)$

$\Rightarrow 6x - y - 5 = 0$

$\because$ It passes through (a, b)

$\Rightarrow 6a - b - 5 = 0$ ..... (1)

Slope of tangent to the circle at (a, b)

$2x + 2y\frac{dy}{dx} - 4 + 2\frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = \frac{2-x}{y+1}$

At (a, b), $\frac{dy}{dx} = \frac{2-a}{b+1} = 6 \Rightarrow 2 - a = 6b + 6$

$\Rightarrow 6b + a + 4 = 0$ ..... (2)

Solving equation (1) and (2) simultaneously $a = \frac{26}{37}$ and $b = -\frac{29}{37}$

$\therefore a + b = -\frac{3}{37}$

80. $f(n) = \lim_{n\to\infty} \sum_{r=n^2}^{(n+3)^2} \frac{1}{\sqrt{r}}$

$\Rightarrow f(n) = \frac{1}{\sqrt{n^2}} + \frac{1}{\sqrt{n^2+1}} + \frac{1}{\sqrt{n^2+2}} + ... + \frac{1}{\sqrt{n^2+6n+9}}$

Total number of terms are 6n + 10

$\frac{6n+10}{\sqrt{n^2+6n+9}} \le f(n) \le \frac{6n+10}{\sqrt{n^2}}$

$\lim_{n\to\infty} \frac{6n+10}{n} = 6$ and $\lim_{n\to\infty} \frac{6n+10}{\sqrt{n^2+6n+9}} = \lim_{n\to\infty} \frac{6+\frac{10}{n}}{\sqrt{1+\frac{6}{n}+\frac{9}{n^2}}} = 6$

$\therefore \lim_{n\to\infty} f(n) = 6$

81. $f(x) = f(10 - x)$

$\Rightarrow f(x)$ is symmetrical about x = 5

$f'(x) = -f'(10 - x)$

Put $x = 5 \Rightarrow f'(5) = 0$

Put $x = -5 \Rightarrow f'(-5) = -f'(15) \Rightarrow f'(15) = -2$

82. Let $f'(x) = k$ where $k \ge 5.4$

$f(x) = kx + c$

Put x = 2

We get $-5 = 2k + c$

$c = -2k - 5$

$f(7) = 7k + c = 5k - 5$
$\because\ k \geq 5.4$
$5k \geq 27$
$5k - 5 \geq 22$

83. $$\sum_{k=1}^{13}\left(1+\tan\left(\frac{k\pi}{12}\right)\tan\left(\frac{k-1}{12}\right)\pi\right)-1=\sum_{k=1}^{13}\frac{\tan\left(\frac{k\pi}{12}-\frac{(k-1)\pi}{12}\right)\left(1+\tan\frac{k\pi}{12}\tan\frac{(k-1)\pi}{12}\right)}{\tan\frac{\pi}{12}}-13$$

$$=\sum_{k=1}^{13}\frac{\tan\frac{k\pi}{12}-\tan\frac{(k-1)\pi}{12}}{\tan\frac{\pi}{12}}-13$$

$$=\sum_{k=1}^{13}\frac{T_k-T_{k-1}}{\tan\frac{\pi}{12}}-13=\frac{T_{13}-T_0}{\tan\frac{\pi}{12}}-13=\frac{\tan\frac{13\pi}{12}}{\tan\frac{\pi}{12}}-13$$

$= 1 - 13 = -12$

84. $f(x) = ae^{x \sin x} + b \cos x$
It is differentiable everywhere

85. $\left|\vec{b}\cdot\left(\vec{a}\times\vec{c}\right)\right| = 35$

$\left|\vec{b}\right|\left|\vec{a}\times\vec{c}\right|\cos\alpha = 35$

$7 \times 5 \times \cos\alpha = 35$
$\cos\alpha = 1 \Rightarrow \alpha = 0^o$
$\therefore\ \vec{b}$ is parallel to $\vec{a}\times\vec{c}$
$\Rightarrow\ \vec{b}$ is perpendicular to $\vec{a}$ and $\vec{b}$ is perpendicular to $\vec{c}$
$\Rightarrow\ \vec{a}\cdot\vec{b} = 0$
If a lies in xy plane then $\vec{a} = a_i\hat{i} + a_2\hat{j}$ and $\vec{a}\cdot\vec{b} = 0$
$\Rightarrow 2a_1 + 3a_2 = 0$
$\Rightarrow\ a_1 = -\frac{3}{2}a_2$

86. $D = (2 - a)(3 - b)$
$Dx = (4b - 15)(a - 2)$
$Dy = 0$, $Dz = a - 2$
If $a \neq 2$ and $b = 3$ then $D = 0$, $Dx \neq 0$, $Dz \neq 0$
$\therefore$ Given system of equation has no solution
$\therefore$ Option (C) is incorrect

87. Median $= \frac{7k + 9k}{2} = 8k$
Mean deviation about median = 11
$$\frac{\sum|x_i - 8k|}{6} = 11$$
$7k + 5k + k + k + 3k + 5k = 66$
$\Rightarrow 22k = 66$
$\Rightarrow k = 3$

88. $e_1 = \sqrt{2}$, $e_2 = \sqrt{2}$ (both are rectangular hyperbolas

$e_5 = 1$ (parabola)

$\frac{1}{e_3^2} + \frac{1}{e_4^2} = 1$ (because both hyperbola are conjugate of each other)

$\therefore \tan\theta = \sqrt{3}$

$\therefore \theta = \frac{\pi}{3}$

89. For the number to be divisible by 3

a + b + c + 5 + 7 + 2 + 4 + 9 must be divisible by 3

$\Rightarrow$ a + b + c + 27 must divisible by 3

$\Rightarrow$ a + b + c must be divisible by 3

$\because$ Distinct digits should be there

Only possibilities for a, b, c are 0, 1, 3, 6, 8

If the possibilities of a, b, c are {0, 1, 8}, {0, 3, 6}, {1, 3, 8}

$\therefore$ Ordered pairs possible = 6 + 6 + 6 = 18

90. Differentiating both sides w.r.t. x

f(sin x) cos x = 2x

Put $x = \frac{\pi}{6}$ we get $f\left(\frac{1}{2}\right) = \frac{2\pi}{3\sqrt{3}}$